WORKED SOLUTION

STEP I 1990, Question 12

Mechanics • Elastic Springs and Energy

STEP I 1990, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

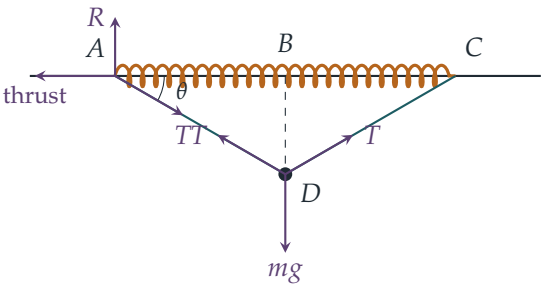
step.maths.org/questions/90-s1-q12

The key idea

Everything here is light except the particle, so the forces at the ends A and C must balance exactly: the horizontal pull of the string on each end equals the push of the compressed spring. That gives the equilibrium condition. For the motion, energy is the natural tool: the only kinetic energy is the particle's, so the particle is at rest precisely when the potential energy returns to its starting value.

Solution

Model. The spring and string are light, the string is inextensible, and the rod is smooth, so the rod can push on the spring only vertically. By symmetry about the vertical through B, the particle D stays directly below B, and $AD = DC = l$. Let T be the tension in each half of the string and θ the angle each half makes with the rod.



Equilibrium

With $\theta = \pi/6$, the half-length of AC is $l\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}l$, so

$$AC = \sqrt{3}\,l < 2l,$$

and the spring is *compressed* by $(2-\sqrt{3})l$. By Hooke's law it pushes outwards on A (and on C) with thrust

$$\frac{\lambda(2-\sqrt{3})l}{2l} = \lambda\Big(1-\frac{\sqrt{3}}{2}\Big).$$

Resolving vertically at D: $2T\sin\frac{\pi}{6} = mg$, so $T = mg$.

Resolving horizontally at A:

Ask yourself Why can we balance forces at A, when nothing is there?

The end of the spring is a light point: any net force on zero mass would give infinite acceleration, so the forces on it must balance at every instant (not just in equilibrium). The rod is smooth, so its reaction R is vertical, and horizontally only the string and the spring act:

$$T\cos\frac{\pi}{6} = \lambda\Big(1-\frac{\sqrt{3}}{2}\Big) \quad\Longrightarrow\quad mg\,\frac{\sqrt{3}}{2} = \lambda\,\frac{2-\sqrt{3}}{2},$$

so

$$mg = \lambda\,\frac{2-\sqrt{3}}{\sqrt{3}} = \lambda\Big(\frac{2}{\sqrt{3}}-1\Big),$$

as required.

The potential energy

Let $AC = 2xl$, so $0 < x \le 1$ while the string is taut. Half of AC is xl, and $AD = l$, so by Pythagoras D is a depth

$$BD = l\sqrt{1-x^2}$$

below the rod. The extension of the spring is $2xl - 2l$, so its elastic energy is $\dfrac{\lambda(2l-2xl)^2}{2\cdot 2l} = \lambda l(1-x)^2$. Taking the rod as the zero level of gravitational potential energy,

$$V(x) = \lambda l(1-x)^2 - mgl\sqrt{1-x^2}.$$

As a consistency check,

$$V'(x) = -2\lambda l(1-x) + \frac{mglx}{\sqrt{1-x^2}},$$

and at $x = \frac{\sqrt{3}}{2}$ (where $\cos\theta = x$, so $\theta = \frac{\pi}{6}$) this vanishes exactly when $mg\sqrt{3} = \lambda(2-\sqrt{3})$, which is the condition we found above.

The fall from B

Initially the particle is at B, so the string lies along the rod, $AC = 2l$ and $x = 1$. Then $V(1) = 0$ and the system starts from rest.

The rod is smooth and the string inextensible, so no energy is lost; and since the spring and string are light, the only kinetic energy is $\frac{1}{2}mv^2$ for the particle. Hence throughout the motion

$$\tfrac{1}{2}mv^2 + V(x) = V(1) = 0.$$

Ask yourself Why does "comes to rest" mean $V(x) = 0$, and not $V'(x) = 0$?

The particle is at rest when $v = 0$, which by energy conservation means $V(x) = 0$. The condition $V'(x) = 0$ describes equilibrium, which is where the particle moves *fastest*, not where it stops. So x satisfies

$$\lambda(1-x)^2 = mg\sqrt{1-x^2}, \qquad \text{that is,} \qquad (1-x)^2 = \left(\frac{2}{\sqrt{3}} - 1\right)\sqrt{1-x^2}.$$

The root $x = 1$ is the starting position. For $x < 1$ we may divide by $\sqrt{1-x} > 0$ and square, which gives a tidier cubic:

$$(1-x)^3 = k^2(1+x), \qquad k = \frac{2}{\sqrt{3}} - 1 \approx 0.15470, \quad k^2 \approx 0.023932.$$

Squaring is safe here because both sides of $(1-x)^{3/2} = k\sqrt{1+x}$ are positive. Let $f(x) = (1-x)^3 - k^2(1+x)$. Then

$$f(0.65) = 0.042875 - 0.039488 \approx +0.0034,$$
$$f(0.66) = 0.039304 - 0.039727 \approx -0.0004,$$

so there is a root between 0.65 and 0.66, very close to 0.66. Since $f'(x) = -3(1-x)^2 - k^2 < 0$, f is strictly decreasing, so this is the only root with $x < 1$. The particle next comes to rest when $x \approx 0.66$ (more precisely $x \approx 0.6589$), with D a depth of about $0.75l$ below the rod.

Checks

- At $\theta = \pi/6$ with $\lambda = l = m = 1$ and $g = \frac{2}{\sqrt{3}} - 1$, the horizontal pull at A and the spring thrust both equal 0.13397, and a numerical derivative of V at $x = \frac{\sqrt{3}}{2}$ is zero (with $V'' > 0$, so the equilibrium is stable).
- Bisection gives the root $x = 0.658861$, and stepping the equation of motion $m\ddot{y} = mg - 2\lambda(1-x)\,y/(lx)$ for the depth y of D (RK4) from rest at $y = 0$ shows the particle first stops at $x = 0.658861$, in agreement.
- The equilibrium value $x = \frac{\sqrt{3}}{2} \approx 0.866$ lies between 0.66 and 1, as it must: the particle oscillates about equilibrium.

Common slips

- Taking the spring to be stretched. Here $AC < 2l$, so it is compressed and pushes A and C apart.

- Setting $V'(x) = 0$ for the rest position. That gives equilibrium, where the speed is greatest.
- Forgetting that the gravitational potential energy is *negative* below the rod, which turns the energy equation into one with no solution in $(0, 1)$.

Beyond STEP

The idea that a massless body must have zero net force on it is a constraint, not a law of motion. Problems like this are handled most cleanly with Lagrangian mechanics, where you write down one energy function in a single coordinate (here x) and the constraint forces never appear. The potential $V(x)$ also gives the period of small oscillations about equilibrium through V''. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats energy methods and small oscillations in this spirit.

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